

CHANG LIN

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Education

Northeastern University, Silicon Valley

Master of Science in Computer Science

San Jose, CA

Sep 2023 – Expected May 2026

University of California, Davis

Davis, CA

Bachelors of Science in Food Science minor in Nutrition Science

Sep 2015 – Jun 2019

Technical Skills

Programming Languages: Python, Java, C/C++, Golang, JavaScript, TypeScript

Frameworks & Tools: Spring Boot, Spring Cloud, Node.js, Express.js, React.js, Maven, JUnit, Mockito, AWS(EC2, S3, Lambda), Docker, Kubernetes, Jenkins, CI/CD pipeline, MySQL, MongoDB, Redis, PostgreSQL, Elasticsearch, RocketMQ, Apache Kafka, RabbitMQ, Git, Linux, Shell scripting, Swagger, Postman

Experience

Convoloo

Jun 2025 – Jul 2025

Software Engineer Intern

Remote, USA

- Developed a full-stack, AI-enabled quiz platform using a 3-tier architecture; engineered a responsive, cross-device compatible frontend with Angular, HTML5, and CSS3, and built scalable backend services in ASP.NET Core (C#) enabling custom quiz generation, real-time analytics, and adaptive learning recommendations
- Integrated Google's Gemini API via Vertex AI using prompt engineering with few-shot learning to drive personalized study paths and multilingual support; crafted structured prompts for effective tool use and context-aware content delivery
- Leveraged Entity Framework Core for SQL Server data operations, optimizing performance with indexing and query caching to reduce API response time by 40%
- Enhanced UI responsiveness and interactivity using Bootstrap and jQuery; implemented frontend performance optimizations including lazy loading and minified assets, reducing initial load time by 35%
- Automated CI/CD pipeline using Docker and GitHub Actions; deployed containerized services to Google Cloud Platform

Guangzhou Panyu District A-list Training Center Co., Ltd

May 2024 – Jul 2024

Software Development Intern

Guangzhou, China

- Developed scalable microservices using Spring Boot and Spring Cloud event-driven architecture to support coupon issuance, redemption, and student management, handling over 10K transactions/min with 99.9% uptime
- Built a Kafka Streams + Apache Avro real-time pipeline to track user activity like class enrollment and coupon usage across partner education platforms, achieving <100ms end-to-end latency and 99.99% message delivery accuracy
- Deployed Redis Cluster with Sentinel to cache frequently accessed coupon and enrollment data, improving coupon redemption latency and reducing backend database load by 20%

Projects

Web Crawler Microservices System

Jun 2025 – Aug 2025

- Designed a distributed web crawler using a microservices architecture, comprising task scheduler, data extraction, indexing, API gateway, and storage services; achieved a crawl rate of 10,000+ pages/min using Node.js
- Developed scalable Node.js services with Express.js, using PM2 for process management, automatic failover, and load balancing; implemented RabbitMQ for reliable task queuing with sub-5ms message latency
- Designed MongoDB schemas using Mongoose for structured storage of over 1TB crawled data, enforcing schema validation and achieving 5ms average query response with 99.99% data consistency
- Optimized performance with Redis Cluster for caching and task queue control, reducing API response time by 20%
- Implemented full-text search with Elasticsearch improving search relevancy by 95% and reducing query latency by 20%
- Deployed the system on AWS EC2 using Docker and Docker Swarm; implemented CI/CD pipelines with Jenkins

Cloud-Native GoPaaS Platform Development

Jan 2025 – Jun 2025

- Designed and built a cloud-native GoPaaS platform using Go-micro v3 and Kubernetes achieving <50ms inter-service latency and 99.95% uptime with features like config management, service discovery, circuit breakers, and rate limiting
- Architected microservices using gRPC and protobuf for efficient inter-service communication; implemented API Gateway with Hystrix circuit breakers, load balancing, and rate limiting, reducing system failure rates by 65% under peak traffic
- Managed scalable Kubernetes clusters with Dockerized services, NGINX Ingress, and custom controllers for traffic routing and job scheduling; enhanced observability and tuning using Prometheus and Grafana, supporting 60% traffic growth

Distributed Key-Value Storage System

Sep 2024 – Dec 2024

- Built a distributed key-value store in Java with Multi-Raft consensus, achieving 99.9% availability, 50K QPS, and <10ms P99 latency for linearizable reads/writes
- Implemented consistent hashing with virtual nodes for 95% balanced data distribution and dynamic shard rebalancing with <1% performance overhead
- Integrated RocksDB (LSM-tree) + in-memory B+ Tree hybrid engine with MVCC snapshot isolation; optimized reads via ReadIndex, FollowerRead, and async Apply batching, reducing latency by 35% and doubling throughput